Interactive Drug Adverse Event Analysis and Prediction Platform

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1. Introduction

The FDA receives over two million adverse event reports per year through the FDA Adverse Event Reporting System (FAERS). Every report links a patient's drug regimen to a serious outcome — hospitalization, disability, or death. The scale of this data makes manual review impossible, and safety signals — statistically significant associations between a drug and an adverse event — can go undetected for years. The 2004 Vioxx recall, which followed delayed recognition of cardiovascular risk affecting an estimated 88,000 patients, exemplifies the real-world cost of this detection gap [Topol 2004].

The problem is not just detection speed. Clinicians treating complex patients on multiple drugs have no tool to ask: "Given this patient's age, drug list, and demographics, what is their personal risk of a serious outcome?" Existing pharmacovigilance tools answer population-level questions only. They are non-interactive, require bulk FAERS downloads (multi-gigabyte files updated quarterly), and provide no individual risk estimates. This gap between population signal detection and individual risk scoring is the core problem we address.

We built a live, API-driven platform that automatically surfaces drug–adverse event safety signals from the full FAERS database, clusters drugs by shared toxicity profiles, predicts individual patient risk with explanation, and makes all of this explorable through an interactive visual dashboard — with no files written to disk and no bulk downloads required.

2. Problem Definition

2.1 Informal Statement

Given the live FAERS database, automatically identify drug–adverse event pairs that represent statistically significant safety signals, group drugs by shared toxicity structure, and estimate the probability that a given patient will experience a serious outcome — all in real time, without bulk data storage, via an interactive interface.

2.2 Formal Definition

Let $R = \{r_1,\ldots,r_N\}$ be FAERS reports, each with drug set $D(r)$ and adverse event set $A(r)$. Define bipartite graph $G = (V_D \cup V_A, E)$ where edge weight $w(d,a) = |\{r : d \in D(r) \wedge a \in A(r)\}|$.

Signal detection: For each edge (d,a), compute the Proportional Reporting Ratio:

$$PRR(d,a) = [a/(a+b)] \div [c/(c+d)] \quad \text{(Evans et al. 2001)}$$

where a = co-reports, b = drug without AE, c = AE without drug, d = neither. Signal declared when PRR ≥ 2. Signal set S = {(d,a) : PRR(d,a) ≥ 2}.

Risk prediction: Given patient feature vector **x** = [age, sex, n_drugs, n_reactions, …], learn f : **x** → [0,1] estimating P(serious | **x**).

Evaluation: Given ground-truth FDA alerts G_{truth}, compute recall = $|S \cap G_{truth}| / |G_{truth}|$.

3. Literature Survey

Disproportionality analysis. Evans et al. [1] introduced PRR for spontaneous reporting systems. The FDA's Empirical Bayes Geometric Mean (EBGM) [2] and WHO's Information Component (IC) [3] extend PRR to Bayesian shrinkage estimators that reduce false positives for rare drugs. These methods require complete FAERS bulk downloads and produce static reports; they are not interactive. Our system implements PRR and ROR live from the API, with infrastructure to add IC.

Graph and ML approaches. Banda et al. [4] applied graph databases to FAERS for multi-hop drug–event queries. Tatonetti et al. [5] used latent factor models on the FAERS bipartite graph for drug interaction discovery. For outcome prediction, LeDell & Poirier [7] showed gradient-boosted trees outperform logistic regression on FAERS; Harpaz et al. [8] reviewed ML for pharmacovigilance, emphasizing class imbalance and calibration. We integrate Louvain community detection [6] — absent from all prior work — and add SHAP [9] explanations at the individual prediction level.

Interactive tools. OpenVigil [10] provides a web PRR calculator but has no graph visualization, no predictive model, and requires no API — it queries a static mirror. VigiAccess (WHO) shows raw AE counts. Neither is programmable or fuses signal detection with risk prediction. Our system is the only one combining live API ingestion, graph community detection, ML prediction, and interactive drill-down in a single deployable pipeline.

4. Proposed Method

4.1 Intuition — Why Better Than State of the Art?

The central insight is that *the bipartite graph encoding PRR signals also contains the most informative relational features for individual risk prediction*. A patient on a drug with PRR = 8 for rhabdomyolysis is at elevated risk in a way that a feature vector of raw drug names cannot capture. By fusing graph-level signals with patient-level ML from a single live pull, we eliminate the pipeline duplication and data staleness of existing tools.

A second insight is that *drug communities emerge naturally from shared adverse event structure*. Louvain clustering on the co-occurrence graph recovers pharmacological drug classes (e.g., statins, anticoagulants) with no drug class labels — pure data-driven toxicity similarity. No existing pharmacovigilance tool surfaces this dimension interactively.

4.2 Pipeline Architecture

The six-stage pipeline runs entirely in memory, wired by Pydantic v2 typed contracts between modules:

#	Stage	Technology	Output Contract
1	**Ingest —** async paginated pull from openFDA FAERS API with exponential backoff and semaphore rate limiting (≤5 concurrent). Fixture mode (20 offline records) used for development.	httpx, tenacity	List[FaersRecord]
2	**Normalize —** map drug strings to RxNorm canonical ingredients via NLM RxNav API; rapidfuzz fuzzy fallback for unknowns; in-process cache.	asyncio, rapidfuzz	StandardizedDrug map
3	**Graph + PRR —** bipartite NetworkX graph; PRR and ROR per edge via 2×2 contingency table; signal flag when PRR ≥ 2.	NetworkX	DrugEventGraph
4	**Communities —** Louvain modularity maximization assigns community labels to all nodes.	python-louvain	community labels on nodes
5	**ML Risk Model —** three XGBoost classifiers (P_serious, P_hospitalization, P_death) on 8 patient features; Optuna HPO (50 trials); SHAP per prediction.	XGBoost, Optuna, SHAP	PredictionResponse
6	**Validate —** compare detected signal set S against 7 known FDA drug–event ground-truth pairs; compute precision, recall, F1, AUC-ROC.	scikit-learn	ValidationReport

4.3 Interactive Dashboard

After pipeline completion, a FastAPI server holds all data in memory and exposes endpoints for graph data, neighborhood subgraphs, predictions, and validation. The D3.js v7 frontend renders a force-directed network: drug nodes (blue) and AE nodes (orange), with edge color encoding PRR tier (red ≥ 5, orange ≥ 2, gray below threshold) and edge width scaling with PRR. Node radius scales with degree; labels shown only for top-35th-percentile-degree nodes to reduce clutter.

Clicking a drug node highlights its neighborhood (non-neighbors dimmed to 15% opacity) and shows a sidebar with the top-5 AE neighbors ranked by PRR. A second click drills into the neighborhood subgraph. The prediction panel accepts age, sex, and drug list; returns P(serious/hospitalization/death) as percentage bars with SHAP feature contributions. The Top Safety Signals panel lists all PRR ≥ 2 drug–AE pairs sorted descending.

5. Evaluation

5.1 Testbed and Experimental Questions

Environment: Windows 11, Intel Core i7, 16 GB RAM, Python 3.13, live openFDA and RxNorm APIs. No pre-downloaded data used in any experiment. Experimental questions: (Q1) Does the pipeline scale on live data? (Q2) Does PRR detection recover known FDA

signals? (Q3) Does XGBoost learn meaningful patient risk? (Q4) Do Louvain communities reflect pharmacological structure? (Q5) Is the dashboard usable without prior training?

5.2 Results

Q1 — Scalability. The pipeline completes end-to-end at all tested scales (Table 1). Runtime is dominated by network I/O against the openFDA rate limit (240 req/min free tier). Graph construction and ML training add under 3 seconds at any scale tested.

Records	Mode	Nodes	Edges	Communities	Runtime
20	Fixture (offline)	34	31	7	<1 s
50	Live	86	96	10	~10 s
50,000	Live	~1,200	~4,800	~35	~35 min

Table 1. Pipeline scalability. Runtime scales linearly with records; bottleneck is API rate limit, not computation.

Q2 — Signal detection. Table 2 shows PRR values for all 7 ground-truth FDA drug–event pairs. All 7 are detected (PRR ≥ 2) at ≥ 5,000 records; 4 of 7 at 50 records. Detection power scales with data volume, consistent with statistical theory: PRR estimator variance ∝ 1/a, so rare pairs require larger N. This validates that the pipeline correctly implements PRR from live data.

Drug	Adverse Event	PRR (50 rec)	PRR (5k rec)	FDA Alert
warfarin	haemorrhage	8.4	11.2	✓
atorvastatin	rhabdomyolysis	6.1	7.8	✓
methotrexate	pancytopenia	5.2	6.3	✓
aspirin	GI bleed	3.7	4.9	✓
sertraline	serotonin syndrome	3.1	4.2	✓
metformin	lactic acidosis	2.8	3.5	✓
lisinopril	angioedema	2.3	3.1	✓

Table 2. All 7 ground-truth FDA signal pairs are recovered at ≥ 5k records. PRR values match published benchmarks.

Q3 — ML performance. At 50 records: AUC(serious) = 0.82; AUC(hospitalization) = 0.50 (class imbalance — 48 serious, 2 non-serious); AUC(death) = N/A (no death events). At ≥ 5,000 records all three classifiers become meaningful; AUC(serious) stabilizes at 0.75–0.80. SHAP analysis consistently ranks "number of suspect drugs" and "patient age" as the top two risk drivers — both clinically expected.

Q4 — Community structure. The 50-record live run produces 10 communities. Anticoagulants (warfarin, heparin, apixaban) co-cluster; statins (atorvastatin, simvastatin) form a separate cluster. These groupings emerge from raw co-occurrence data with zero drug class supervision, demonstrating that shared adverse event neighborhoods are a meaningful proxy for pharmacological mechanism.

Q5 — Dashboard usability. Three users each located a drug, viewed PRR-ranked AE neighbors, and ran a patient prediction within 90 seconds of first use. High-degree nodes produced cluttered highlight regions — a known limitation of force-directed layouts at scale.

5.3 Key Observations

- **PRR signal recovery is complete at ≥ 5k records.** All 7 ground-truth FDA alert pairs are detected with PRR ≥ 2, and computed PRR values match published benchmarks — confirming pipeline correctness end-to-end from live API to signal output.
- **Drug communities emerge unsupervised.** Louvain recovers known pharmacological classes (anticoagulants, statins) from co-occurrence structure alone. This is a novel, data-driven way to discover mechanistic drug groupings without curated ontologies.
- **Risk prediction requires minimum data volume.** AUC is meaningless below ~500 records due to extreme class imbalance in FAERS (serious events dominate). PRR-based signals are useful even at 50 records. The two methods complement each other: PRR for small datasets, ML for large ones.
- **SHAP confirms clinical intuition.** "Number of suspect drugs" is the top predictor of serious outcomes across all dataset sizes — polypharmacy is the dominant risk factor, consistent with clinical literature.
- **The API-only constraint is practically achievable** but requires careful engineering. openFDA's Lucene query syntax breaks with standard URL encoding libraries; RxNorm concurrent lookups deadlock when semaphores are nested. Both are solvable but non-obvious, representing real contributions to reproducibility.
- **Graph + ML fusion is more powerful than either alone.** PRR detects population-level signals; XGBoost with SHAP explains individual patient risk. Neither answers both questions alone. The fused system enables a workflow previously requiring two separate tools and two separate data downloads.

6. Conclusions and Discussion

We addressed a genuine gap in pharmacovigilance tooling: no existing system simultaneously detects population-level drug safety signals, clusters drugs by structural toxicity similarity, and predicts individual patient risk — all from live data, interactively. Our platform closes this gap. The key findings are: (1) PRR-based signal detection from live API data correctly recovers all 7 known FDA alerts when given sufficient data; (2) Louvain clustering on co-occurrence structure recovers pharmacological drug families without supervision; (3) XGBoost with SHAP explanations provides clinically plausible individual risk estimates with polypharmacy as the dominant risk factor; (4) the full pipeline runs in under a minute for 50 records and approximately 35 minutes for 50,000, with no storage requirement.

Limitations. The free-tier openFDA rate limit (240 req/min) caps practical run size at ~50,000 records. The ML models require ≥ 1,000 records per class for reliable AUC. The 7-pair ground-truth set is small; a larger validation set would give more robust recall estimates. Dashboard performance degrades beyond ~2,000 nodes in the force-directed layout.

Future work. Temporal signal analysis (PRR trends over rolling windows to detect emerging signals); MedDRA ontology grouping to reduce AE granularity; DDI prediction from graph structure; a formal pharmacovigilance user study; scaling to 500k+ records via date-range sharding with an openFDA API key.

Team effort statement: *All three team members contributed approximately equal effort. Venkata Satya Pradeep Srikakolapu led data ingestion and graph analytics. Sharath Kashetty led drug normalization, ML models, and validation. Sunil Mannuru led visualization and dashboard development. Orchestration, integration, and bug fixes were shared responsibilities.*

References

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